Behavioral Statistics: Theory and Application

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Preface

0.1 You can learn statistics

Many textbooks in mathematics begin with a useful high level overview of the content of the book. We'll get to that in a moment. I want to start this one off a little differently: *You can learn statistics.* I believe that you can learn statistics and I'm happy you have opened this book to do it. Statistics is a wonderful field, and I am excited for you to learn it. I believe in you.

0.2 Why I wrote this book

I wrote this book because I'm excited about statistics (really excited; who writes a textbook for fun?), and I want other people to be excited about statistics too.

During my years as an academic, I became very comfortable with the applied end of statistics. I nevertheless found myself approaching statistics like a recipe book, e.g., if you have this type of data, it should be analyzed in this way. I did eventually pick up semi-mathematical "explanations" for why certain analyses were appropriate, and although these explanations were accurate (in a sense) they were sometimes quite amorphous. I have grown to appreciate the concrete and exact nature of mathematics - particularly the field of mathematical statistics. This book is therefore meant to teach some deeper mathematical theory *in addition to* the application of statistics. My hope is that some curious student (like myself long ago) can have their curiosity assuaged.

0.3 What content to expect in this book

To that end, this book will review many concepts learned in an undergraduate behavioral statistics course, but with more mathematical emphasis. It will proceed in three parts:

1. Chapters 1, 2, and 3 give a background on probability and probability distributions. Within behavioral statistics courses, we often discuss a vaguely defined "population" of study. These three chapters define more concretely what we mean when we refer to a population or populations. Hint: Populations can be thought of as random variables and their corresponding **probability distributions**. We'll dig into this later.

2. Chapter 4 is our first foray into statistical inference. While Chapters 1, 2, and 3 tell us about populations, this chapter will begin our journey of **inferring** some basic things about these probability distributions. We will learn about the **Law of Large Numbers** and the amazing **Central Limit Theorem**.

3. Chapter 5 (and the chapters thereafter) will introduce us to hypothesis testing - the cornerstone of statistical inference in the behavioral sciences. We will cover several statistical techniques such as z-tests, t-tests, correlation, chi-square tests, and ANOVA.

Thank you for reading!

0.4 What you'll need to know to get the most out of this book

The most important "thing" you need to read this book is excitement. I would not describe myself personally as a mathematical prodigy in any way. I'm just excited about the subject, and that provides me with the extra boost to understand the difficult stuff. The second thing you'll need is an okay grasp of algebra. If you are rusty, I suspect enough will come back to you to understand everything in this book. I'll try to give gentle reviews where I can throughout. There is lots of content online (like this book) that would offer an awesome review of algebra, so if you are confused and I don't provide a review, you could almost certainly find something to get you up to speed.

0.5 Future editions of this book

This book is a working document. It was created based on some hand written notes I developed whilst teaching my colleagues. I am *still* teaching my colleagues and I shall try to add chapters in pace with that endeavor. I also plan to work in practice problems and tutorials on R sometime in the future. Stay tuned.

Chapter 1

Set Theory and Probability

In the course of research, behavioral scientists will often take samples of human behavior, e.g., a survey of a group of people, measurements of reaction time, etc. These samples will take some *outcome*, and most of us understand and have observed the inherent randomness of this act. That is to say, you might take multiple samples in a carefully controlled circumstance and get different results every time.

Statisticians often refer to an analogous circumstance when talking about probability. Suppose we are able to describe all the possible outcomes of our sampling procedure. We take samples over and over again in the same circumstances, and each time the outcome may change. This process is commonly referred to as a **random experiment**. In order to familiarize you with some introductory concepts around probability, we're going to start within the context of a random experiment. First, we'll learn about sets and set theory in order to describe our **sample space**. This be useful in describing the form that outcomes of a random experiment may take. Then we'll more concretely define the randomness of our experiment by talking about **probability** - the quantification of randomness.

1.1 Sets and Sample Spaces

Before we get into it, we aren't going to learn set theory in it's entirety. That would be overly (and hilariously) ambitious. What we are going to learn will make understanding later sections easier, get your brain warmed up, and make it a bit easier to understand set notation in other circumstances. The first step in conducting a random experiment is clearly defining all the possible outcomes of your sampling procedure. This set of outcomes is referred to as the **sample space**. For example, suppose we have five different possible outcomes in our sample space - A, B, C, D, and E. For simplicity sake, Let's refer to this set of outcomes as S (S for Sample!). Then we would describe our set like this:

$$S = \{A, B, C, D, E\}$$

That is to say we'll describe S with curly braces surrounding a list of *distinct* elements in the set. The set S is an example of a **discrete** sample space - meaning it either contains a finite number of elements or it is *countable*. In this case, the sample space has a finite number of elements (five to be exact), so we can call it discrete.

Describing and identifying a discrete sample space that is countable but *infinite* (has an infinite number of elements) is much more difficult. My hope is you can gain a small sense for it from my explanation here. We'll see examples of it later, so it's worth a brief consideration. Specifically, a discrete sample space can also contain an *infinite* number of elements - so long as those elements can be lined up one-to-one with the natural numbers (e.g., 1, 2, 3, 4, …). That is, the elements can be "counted", albeit infinitely and forever. This is probably not the most intuitive definition for people who don't think about mathematics every day (it certainly wasn't for me). In a more reachable sense, a discrete sample space has elements with nothing in between them. In S above, there is nothing in between A and B, for example. If your discrete set had an infinite number of elements, you could order them by some criteria and line them up with the natural numbers. After this alignment, you could find nothing between the first and second elements (or any given sequential pair for that matter). You could just keep counting without worrying that you missed something.

In addition to the discrete sample spaces above, we may also find ourselves sampling from a **continuous** sample space, for example, all the values between and including 0 and 1 (e.g., $0 \leq x \leq 1$). Since you can't enumerate all the values of a continuous sample space as we did with S (people have "tried"; we'll talk about one such example later), we use a different form of notation. Suppose we have a sample space consisting of all the values between and including 0 and 1. Like the discrete case, we'll give the set a name. In this case R seems like a nice name (R for Research!), and we will describe R as

$$R = \{x \,|\, 0 \leq x \leq 1\}$$

The inclusion criteria for the set is written after the | in the curly braces. The | can be read loosely as meaning "where", so reading the whole thing in the curly braces left to right you get x *where* $0 \leq x \leq 1$. Therefore, our set is all the x's from 0 to 1. Note that I will drop the x and | when it is clear what variable I am referencing. As an aside, continuous sample spaces can be found everywhere in the field of behavioral statistics. For reasons that will become apparent later, we shall nonetheless approach these spaces with slightly less mathematical depth.

Now, suppose we conduct a random experiment with our previously described sample space $S = \{A, B, C, D, E\}$. The *outcome* of our experiment is the single item from the sample space identified via our random process (e.g., a random draw with eyes closed, a dart board with a blind fold; etc.). In practice, we will frequently talk about outcomes in terms of *events*. Events are described by a *subset* of distinct elements from the sample space. That is, they may be one or more outcomes combined. For example, we might describe an event as $C = \{A, B, C\}$, and if our outcome (a single item) lands in the that subset (it is either A, B, or C), we shall say the event "occurred". For the sake of brevity, and positivity, I will frequently refer to the occurrence of an event as a "success" (congratulations all around!). You can indicate a set is a subset by:

$$C \subset S$$

In this case, we are saying C is a subset of S. We might also, for example say that our event is a single element/outcome $C_1 = \{A\}$ or the entire sample space $C_2 = \{A, B, C, D, E\}$. We can be as creative as we would like to be, so long as the event is in our sample space.

Indeed, in practice we might want to get a little more creative in terms of the outcomes and events associated with our random experiment. For example, we might be interested in the combination of events, the overlap in events, or some other scenario. The next few examples will describe (at a basic level) how we might "combine" events in different ways.

Example 1.1. Suppose we have two events $C_1 = \{A, B, C\}$ and $C_2 = \{B, C, D\}$, and we are interested in whether or not our outcome lands in C_1 *or* C_2. Let us first consider the circumstances where our experiment would result in a success. We might reason that if the outcome of our random process was an A, B, C, or D, then our outcome will have landed in one of the two sets. Specifically, if our outcome were an A, then it has landed in C_1; a B or C, then it has landed in C_1 and C_2; and a D, then it has landed in C_2. That is to say, we could take the unique outcomes of either set and combine them into a new set. If our outcome lands in that new subset, we can be confident that C_1 or C_2 occurred. There's a handy symbol that indicates this operation:

$$C_1 \cup C_2 = \{A, B, C\} \ \cup \ \{B, C, D\} = \{A, B, C, D\}$$

This *union operator* ($\cup$) takes the unique elements of both sets and puts them into a set of their own. In this circumstance it is used synonymously with *or*. If the event of interest is our outcome landing in C_1 OR C_2, then we would be successful if the event landed in the union of the two sets, e.g., all the unique elements from both sets.

Example 1.2. Here's another example, suppose we are interested in whether or not our outcome falls into $C_1 = \{A\}$ or $C_2 = \{B\}$. Then our event would occur if the result of our random process were an A or B. Using our newfangled union operator, we can write the union of these two sets as

$$\{A\} \cup \{B\} = \{A, B\}$$

Thus, the union operator helps us determine the conditions for success when we are interested in the occurrence of one event OR another. Let us now turn our attention to another equally important circumstance.

Example 1.3. Suppose we have our two previously described events - $C_1 = \{A, B, C\}$ and $C_2 = \{B, C, D\}$ - each a subset of S. Let's further say that we are interested in whether or not both events occur as a result of our random experiment. That is, we shall be successful if our outcome falls in both C_1 AND C_2. We are therefore interested in the set of *shared* elements of our two events. Specifically, just by eyeballing the sets above, B and C are shared by the two events C_1 and C_2, but if we were unfortunate enough to draw a D or an A, game over. The *intersect operator* will help us signify this process:

$$C_1 \cap C_2 = \{A, B, C\} \cap \{B, C, D\} = \{B, C\}$$

Unlike $\cup$, the intersect operator represents **and**. It tells us the conditions for success if we would like *both* events to occur (a.k.a., our outcome landing in both events). This operator can create some interesting conundrums, so it's worthwhile to do another example or two.

Example 1.4. This time let's define our events to be $C_1 = \{A, B\}$ and $C_2 = \{C, D\}$. Now we're interested in whether or not our outcome falls in C_1 and C_2 (What do you think is going to happen here?):

$$C_1 \cap C_2 = \{A, B\} \cap \{C, D\} = \{\}$$

Well, you may have guessed it: Our two sets had no overlap. You couldn't possibly draw a single element that landed in *both* the sets, because the sets share no elements. As a result, we get a very special set - a *null set* - $\{\}$ - which is a set with zero elements. In this circumstance, when the intersect of sets results in a null set (a.k.a. they have no overlap), we shall refer to them as being **mutually exclusive** - because the two sets do not share elements. In terms of probability, mutually exclusive sets can be much easier to work with. We'll talk about this soon.

Now you have some understanding of the two main set operators we'll need - the union ($\cup$) and the intersect ($\cap$). We shall need to know a little bit more about the null set, so let's take a slightly deeper look.

Example 1.5. It's interesting to see how a null set works with other sets that have elements. For example:

$$\{A, B, C, D, E\} \cap \{\} = \{\}$$

A set with elements intersected with a null set is a null set. A set with something in it can share nothing with a set that has no elements at all. This is very similar to multiplying an integer by zero. In fact, when you first start writing set notation, you might be inclined to use arithmetic operators with sets, e.g., $\{A, B, C\} - \{A, B\} = \{C\}$ as opposed to $\{A, B, C\} \cap \{A, B\} = \{C\}$. The null set and intersect gives us a handy way to multiply a set by zero without using $\times$ or 0.

What do you suppose happens when we union a null set with something else? Let's try it out:

$$\{A, B, C, D, E\} \cup \{\} = \{A, B, C, D, E\}$$

A set unioned with a null set is itself. A null set has nothing unique to add to our "something set". Although the null set is really cool (in my opinion; and I hope your opinion too), working with it is, metaphorically speaking, like talking to the most boring person in the world. You have nothing in common with them ($\cap$), and they have nothing to add ($\cup$). And, with that, you have learned your first set theory insult.

Up to this point, we've learned how to describe our sample space, the concept of an outcome and event, how to create new events with union/intersect (or/and) operators, and the fancy null set. That's about all we need. I would like to finish out this brief section on sets with a discussion of the *complement* of a subset and a brief treatment of how the previously mentioned operators work in the continuous case.

The complement of a subset describes all the elements of a sample space *not* in that subset. For the sake of consistency, we'll keep using our sample space S (**??**), and let's work with $Y = \{A, B, C\}$, $Y \subset S$. The complement of that subset in the sample space would be $\{D, E\}$. The operation of taking the complement of a subset is indicated by:

$$Y^c = \{A, B, C\}^c = \{D, E\}$$

The complement allows for a wider breadth of operations.

Example 1.6. Let's again consider our sample space S. Suppose we were interested in an event that was NOT in the two subsets $C_1 = \{A, B\}$ or $C_2 = \{C, D\}$. The first thing we would do in this circumstance is maybe consider all the ways our outcome could be unsuccessful (we won't use "fail" here; too negative). It would be unsuccessful if it landed in C_1 or C_2. Recall that the $\cup$ (union) gives us the unique elements in both sets. Let's take these two sets and union them into a set X together.

$$C_1 \cup C_2 = \{A, B, C, D\} = X$$

Now that we have our subset of unsuccessful elements, we just take the complement of that:

$$X^c = \{E\}$$

Of course, you might have known the answer to this all along, but it's nice to see the steps broken down exactly. You could have written it all in one go using parentheses to specify order of operations, e.g.:

$$(C_1 \cup C_2)^c = \{A, B, C, D\}^c = \{E\}$$

Notice how we do the operation in the parentheses before taking the complement.

Example 1.7. Let's try one more example with C_1 and C_2. Let's say we defined a successful outcome as all the things NOT in C_1 and C_2. Notice the key word "and". Our list of unsuccessful items will be different this time because we are referring to the shared (intersect) of both our subsets. Therefore:

$$(C_1 \cap C_2)^c = \{\}^c = \{A, B, C, D, E\}$$

The two sets have no overlap, so their intersect is a null set, and the complement of the null set is all the elements in our sample space.

With a good understanding of some of the basic operators and sets, let's turn to a brief treatment of the continuous sample space. The continuous case is very similar to the discrete case - although it takes some drawing of number lines if you are rusty with inequalities (no problem).

Example 1.8. Suppose we are sampling from a sample space $S = \{x \,|\, 0 \leq x \leq 10\}$ and we are interested in if the outcome lands in $C_1 = \{x \,|\, 0 \leq x \leq 2\}$ OR $C_2 = \{x \,|\, 2 \leq x \leq 5\}$. A successful outcome here would occur if our realized element landed anywhere from zero to five. In other words:

$$\{0 \leq x \leq 2\} \,\cup\, \{2 \leq x \leq 5\} = \{0 \leq x \leq 5\}$$

Let's keep the same continuous sample space and subsets as above, and this time let's look at the intersect of the two sets:

$$\{0 \leq x \leq 2\} \,\cap\, \{2 \leq x \leq 5\} = \{2\}$$

The two sets only overlap at exactly $x = 2$, so their intersect is just a set with the value 2. Let's do a couple more so you can get a sense for it.

Example 1.9. Suppose we have a sample space $Q = \{x \,|\, 5 \leq x \leq 10\}$ and define $C \subset Q$ where $C = \{x \,|\, 5 \leq x < 7\}$. That is to say, C is a subset of Q, and its interval includes the lower side (5) but it doesn't include the upper side (7). What is the complement of C?

$$C^c = \{5 \leq x < 7\}^c = \{7 \leq x \leq 10\}$$

The complement includes 7 up to 10. In the plot below, you can see a visual depiction of this. The top line represents the subset C, and the bottom line represents the complement. The open circle at the end of our subset C indicates that the value 7 is not included in its range. The elements of C are strictly less than 7. This is in contrast to the complement line below - which has a solid black dot indicating 7's inclusion due to its exclusion from C. Thus the complement of C ranges from 7 to 10.

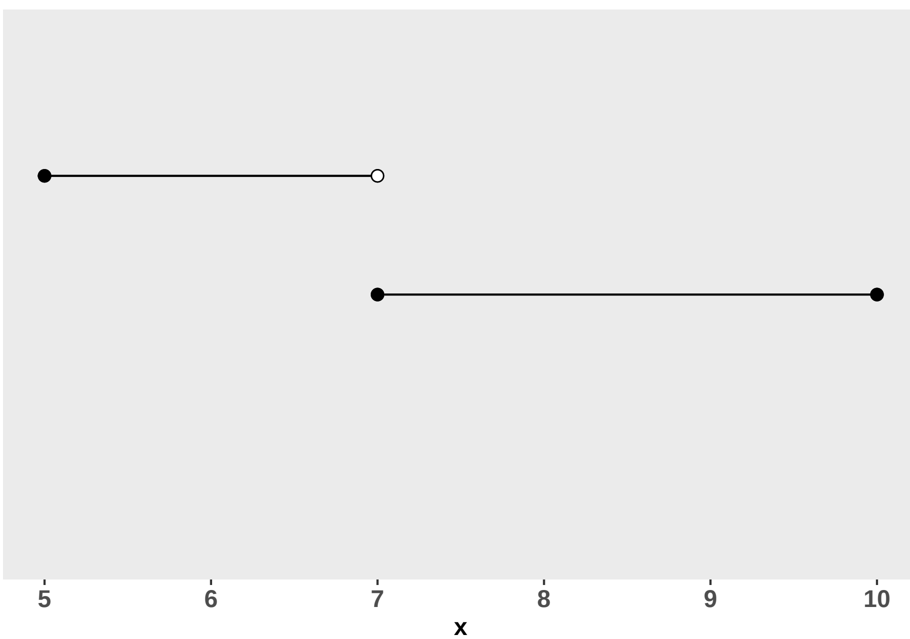

Figure 1.1: Plotting a Complement

Example 1.10. One more comment is worth a mention with regards to continuous sample spaces. Suppose we have two subsets of our space Q that don't overlap in any way. For example, take $C_1 = \{x \,|\, 5 \leq x < 7\}$ and $C_2 = \{x \,|\, 9 \leq x \leq 10\}$. Then you would simply write their union as:

$$\{\, 5 \leq x < 7\} \cup \{\, 9 \leq x \leq 10\}$$

There is no simplified way to write this in set notation. Note the intersect of these two sets would be $\{\}$ - a null set.

1.1.1 Repeated Samples

The previous introduction to sets was dedicated to a random experiment where we only draw a single element from our sample space. In practice, researchers and statisticians are often sampling/conducting a random experiment repeatedly. In this section we'll talk about how to describe the sample space of an experiment that involves multiple draws. Note, for the purposes of our learning, we shall only be discussing sampling with *replacement*. That is, we will be talking about circumstances where the 1st, 2nd, 3rd, … , nth outcome gets returned to the sample space prior to the next draw. This is opposed to the circumstance where an outcome of a single sample reduces the size of the sample space each time. For example, suppose we have a sample space, $S = \{A, B, C\}$. On our first experiment, we randomly draw an A. In sampling without replacement, the sampling space for the next draw would be $\{B, C\}$ (or $\{A\}^c$) and in sampling with replacement A would be returned to the space.

To understand how to describe the sample spaces mentioned above, we'll turn to a good-old-fashioned coin toss (the mainstay of every introductory probability book). Suppose we perform a single flip (experiment) from a fair coin. Our sample space would be $S = \{H, T\}$ where H is "heads", and T is "tails". Now let's suppose we flip two counts. The sample space would be all the possible permutations of heads and tails from the two flips. It helps to draw draw a *tree diagram* to get the gist of it.

You can follow the branches of the tree above to see each possible permutation of the two flips. For example, suppose the first flip was heads, you would follow the line at the first node (on the left) up the the "H" (heads) node. Then at the "heads" node, the next flip could either be an "H" (heads) or a "T" (tails), corresponding to the top and second from the top nodes on the right. Going down each branch, top-to-bottom, we get all the possible permutations of our two flips - $\{HH, HT, TH, TT\}$. Our random experiment is now a selection from this new sample space.

Drawing a tree to determine all the permutations of multiple draws, could get quite cumbersome, so it's convenient to have a formula. In the example above, the first flip could take 2 outcomes. The second flip would then result in two additional branches for each of the outcomes of the first flip. In other words, we have $2 \times 2 = 4$ possible permutations for the two coins. Had we flipped a third coin, each of nodes for the second flip would then be accompanied by two branches per node, so the calculation would be $2 \times 2 \times 2 = 8$ permutations. Indeed, given a sample space that has N total elements/outcomes and k repeated samples, all the possible permutations would be given by:

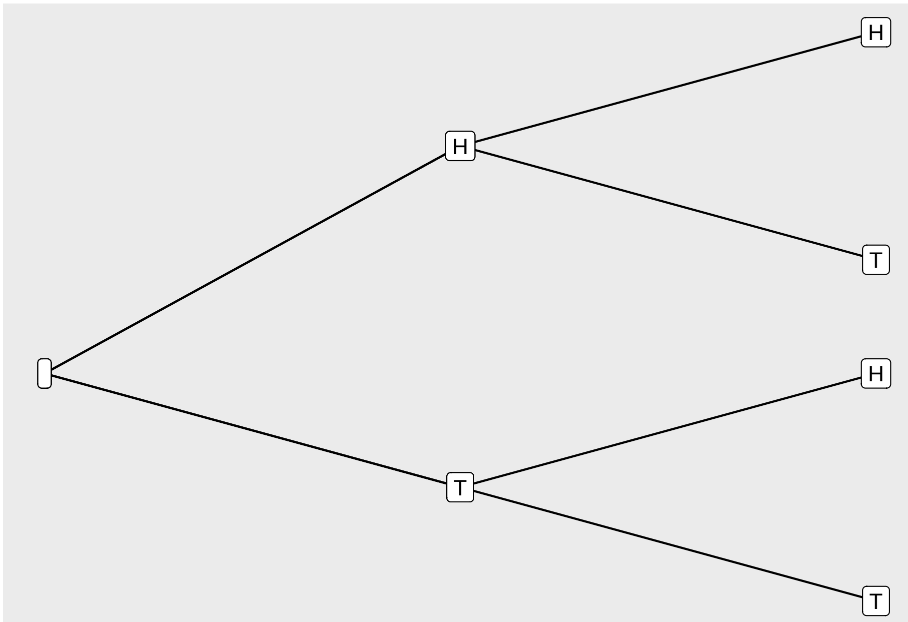

Figure 1.2: Example Tree Diagram for Two Coin Flips

$$N^k$$

Note: If we had sampled from several sample spaces, all with a different number of elements, then then we would just multiply the respective element lengths together.

I will end this section with an important discussion on notation. As you can see from (**??**), the sample space can grow quite rapidly. For example, four coin flips would have a sample space of 16 elements. Consider the two coin sample space - $\{HH, HT, TH, TT\}$. The event of getting a head on the first flip can be described by all the coin flip combinations with heads on the first flip - $\{HH, HT\}$. If our random experiment and outcome fell into this subset, we would indeed have flipped a heads on the first coin. Now consider a circumstance where the sample space is much larger (e.g., 10 coins), it would become quickly untenable to write out all the possible permutations. Instead in some circumstances I may use some short hand like or similar to $\{H\ on\ 1st\ flip\}$, because I don't have the rest of my natural born life to list permutations (and you don't have the rest of your natural born life to read them). Rest assured that these are exactly equivalent, and they shall be associated with the exact same probabilities. Indeed, sometimes the short hand will make it more clear as to what probability we are referring to.

1.2 What is probability?

Many of us have an internal sense of this stuff we call probability. In the crisp fall of the California foothills (which is about the time that I'm writing this section), you can ask a local of Placerville how often the apple orchards get overwhelmed with tourists. More often than not, they will give you some percentage of the time, and this time of the year that percentage will undoubtedly be 100%. You might ask a Sacramento resident how confident they are in their local basketball team winning, and indeed, they may also provide you with something close to a percentage of time (my guess is 15%). The two preceding scenarios are examples of the **relative frequency approach** to probability. In this approach, you can think of probability in the random experiment sense. That is, if we did a random experiment over and over again, we might expect some event to occur a percentage of the time. We shall call this percentage a probability. Within this approach, probabilities can take values anywhere from $0 \leq x \leq 1$. Indeed a "percentage of time" greater than 1 or less than zero would be nonsensical.

In the last section we discussed how we might describe sample spaces, outcomes, and events of random experiments. In this section we'll assign probabilities to them, and we shall adopt the above relative frequency approach interpretation. Suppose we have some event C, then the **probability set operator** - $P(C)$ - takes C, and based on a set of rules, returns a probability for that event. The set of rules for the assignment of probability is referred to as a **probability distribution**. We'll talk about probability distributions in the next chapter, but for now we'll just focus on probability and the probability set operator. Calculating probabilities for the continuous sample space can be tricky, so we'll stick with the discrete case and talk about the continuous later.

Example 1.11. Consider for a moment the circumstance at the end of last section: Flipping two fair coins. If you recall, our sample space was $S = \{HH, HT, TH, TT\}$. Suppose the probability set operator assigns a probability of 1/4 to each of the outcomes in this sample space. The probability, for example, of the event $C = \{HH\}$ would therefore be

$$P(C) = P(\{HH\}) = 1/4$$

The probability set operator takes the set C and assigns to it a probability.

Within the relative frequency approach, the assigned probability of 1/4 means that over many many random experiments, we might see that about 25% of the outcomes are HH. When I have taught this to my colleagues and students, many will point out that they have flipped two fair coins a large number of times and didn't get exactly 25% of outcomes landing HH (budding statisticians!). That is indeed possible. When I talk about "many" trials, I am referring to an exorbitantly large number of trials (e.g., infinitely large numbers of flips).

Indeed, if we were to flip two truly fair coins until the end of time, a total of 1/4 of those flips would be HH.

Example 1.12. While you wrap your mind around that, let's now turn our attention to a more informative example. We'll stick with fair coins for now. Suppose we wanted to know the probability of getting a heads on the first flip - $C = \{HH, HT\}$ - how does the probability set operator go about assigning probabilities to an event with more than one outcome? Well, lucky for us, there is the **law of addition** for probability. For two events A and B for which $A \cap B = \{\}$ (e.g,. they are mutually exclusive):

$$P(A \cup B) = P(A) + P(B)$$

Put in plain language: The probability of our outcome landing in A or B ($\cup$) is equal to the probability of A plus the probability of B. This law helps calculate the probability of success if we want our outcome to land in one event OR another event. In this case, how does it apply to our single event C above? Well, recall from our earlier discussion, we said that an event contains a set of *distinct* elements from the sample space. Because of this, every event can just be written as a union of mutually exclusive events. In our case,

$$C = \{HH, HT\} = \{HH\} \cup \{HT\}$$

And with this in mind, we can apply the law of addition:

$$P(\{HH\} \cup \{HT\}) = P(\{HH\}) + P(\{HT\}) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

Of course, in practice, you don't have to write an event as a union of distinct events or outcomes before applying the law of addition. You can simply sum up the respective probability of each outcome in the event.

Example 1.13. For example, suppose we were interested in the probability of at least one head. In this case, a successful outcome would be HT, TH, or HH, whereas TT would be unsuccessful. Thus we are considering the probability of the event - $C = \{HT, TH, HH\}$, and with the law of addition:

$$P(\{HT, TH, HH\}) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

The law of addition makes things easier in multiple ways, and it implies another (easier!) way to calculate the probability above. To see this, we shall start with a handy property of the probability set function. Namely, for some sample space S:

$$P(S) = 1$$

In English, this means that the probability our outcome lands in the sample space is equal to 1. Of course this is true! The sample space is - by definition - the collection of all the possible outcomes of our random experiment. As I said, this property makes some computations easier.

Example 1.14. For example, suppose we are interested in the probability of getting at least one head (as above). In this case, we don't need to go through the rigmarole of enumerating all the combinations with an H. We know there's exactly one combination with no heads - TT - and the rest of the outcomes (the complement of TT; $\{TT\}^c$) have at least one H. So, let's say that the event C contains the outcomes with at least one head, and let's also say that S is our full sample space:

Observe that:

$$C \cup \{TT\} = S$$

Look back at the definition of our two-coin flip space, and you'll see why I'm right about this. So then we can do this:

$$\begin{aligned} P(C \cup \{TT\}) = P(S) = 1 && (1.1) \\ P(C) + P(\{TT\}) = 1 && (1.2) \\ P(C) + P(\{TT\}) - P(\{TT\}) = 1 - P(\{TT\}) && (1.3) \\ P(C) = 1 - P(\{TT\}) && (1.4) \\ P(C) = 1 - \frac{1}{4} = \frac{3}{4} && (1.5) \end{aligned}$$

Thus, without listing out all the permutations of heads and tails for our two coins, we've calculated what we need. That is to say, sometimes it's much easier to calculate the probability of the complement (in the case above TT) of our event of interest and subtract the probability from 1. More formally, to calculate the probability of some event C, we could take its complement C^c. Then

$$P(C) = 1 - P(C^c)$$

Example 1.15. Let's work in another example of this just to hammer this concept home. Let's say we have a sample space $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the probability set operator assigns a probability of 1/11 to each outcome in this sample space. We want to know the probability that our draw will land in the event $C = \{3 \leq X\}$ - an outcome of 3 or more. You might be able to do this in your head. I'm personally too lazy to write out the law of addition for C, so we'll use (**??**). First observe that

$$C^c = \{0, 1, 2\} P(C^c) = \frac{1}{11} + \frac{1}{11} + \frac{1}{11} = \frac{3}{11}$$

And therefore

$$P(C) = 1 - P(C^c) \tag{1.6}$$
$$P(C) = 1 - \frac{3}{11} \tag{1.7}$$
$$P(C) = \frac{8}{11} \tag{1.8}$$

Much easier!

Okay, I have to stop the narrative now to tell you I have done a great misdeed. For the sake of simplicity I've left out an important part of the law of addition (I know: $P(Hell) > 0$). Now it's time for the truth, and I'll show you what I've done and what we need to add.

Example 1.16. Let's define our sample space to be $S = \{1, 2, 3\}$, and suppose each outcome has a probability of 1/3. Define two events $C_1 = \{1, 2\}$ and $C_2 = \{1, 2\}$. We would like to know the probability of our outcome landing in C_1 or C_2. This sounds like a job for the law of addition. However, remember our current version of the law only works for mutually exclusive sets, but $C_1 \cap C_2 = \{2\}$.

We could always get around the conundrum describe above by executing the set operation before applying the probability set operator. Then with that new set, we could calculate the probability as in (**??**), e.g., using the first law of addition I described for mutually exclusive events. For example:

Observe that
$$C_1 \cup C_2 = \{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$$
And therefore the probability of C_1 or C_2 is

$$P(\{1, 2, 3\}) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

You can see that works pretty well. If we want to use the law of addition, we will nevertheless need to make a small addition. Specifically, we shall have to subtract out the intersect ($\cap$) between our two sets. That is, given two events A and B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This law says that if we were to simply sum the probability of each event, we would double count the overlap between sets A and B ($\cap$) . In the case of our example, we would double count the probability of $C_1 \cap C_2 = 2$ twice. If you don't believe me, look at this. Consider C_1 and C_2 as defined above

$$P(C_1) = P(\{1,2\}) = P(1) + P(2)P(C_2) = P(\{2,3\}) = P(2) + P(3)$$

So if we applied our first version of the law of addition, then

$$\begin{aligned} P(C_1 \cup C_2) &= P(C_1) + P(C_2) && (1.9)\\ &= P(1) + P(2) + P(2) + P(3) && (1.10)\\ &= P(1) + 2P(2) + P(1) && (1.11) \end{aligned}$$

We have one too many $P(2)$s! Indeed, if we counted the probability of 2 twice, we would end up with a nonsensical value of 4/3 (e.g., $1/3 + 2 \times 1/3 + 1/3$). So let's put it all together using the second (more complete) law of addition. Recall that we assigned a probability of 1/3 to each element in our sample space

$$\begin{aligned} P(C_1 \cup C_2) &= P(C_1) + P(C_2) - P(C_1 \cap C_2) && (1.12)\\ P(C_1 \cup C_2) &= P(\{1,2\}) + P(\{2,3\}) - P(\{2\}) && (1.13)\\ P(C_1 \cup C_2) &= \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1 && (1.14) \end{aligned}$$

Fixed!

Example 1.17. Using the full version of the law of addition helps to make clear why we can drop the last term for mutually exclusive sets (**??**). Suppose we have two mutually exclusive events A and B, each a subset of some sample space S. Then, because they are mutually exclusive, $A \cap B = \{\}$. And, indeed, the probability of the null set - $P(\{\})$ - is zero. You can't draw nothing on a random experiment. So in the case of mutually exclusive events

$$P(A \cup B) = P(A) + P(B) - 0$$

What we just learned will serve as a nice initial step into probability. We shall close this session by further emphasizing that $P(A \cup B)$ can be thought of as the probability that our outcome lands in A OR B. On the other hand, $P(A \cap B)$ can be thought of as the probability our outcome is an element shared by A AND B. Indeed, although we can use our set operations and the probability set operator to calculate the ladder, there is another method that we will discuss next.

1.3 Conditional Probability and Independence

Having learned a little bit about probability and the law of addition, we will turn to a slightly more complicated topic - conditional probability. **Conditional Probability** describes the probability of an event given the occurrence of another event. For example, suppose you are on a game show where you select one of four doors. Further suppose that a prize is equally likely to be behind each door. That is, the probability set function assigns a probability of 1/4 to each door. Right before you select a door, the host yells, "stop!", and then it is revealed that the prize is behind doors 1, 2, or 3. That is some fortuitous news! Common sense says that the probability our prize lands behind door number 4 is now *zero*, and we should constrain our guess to doors 1, 2, and 3. That is to say, the conditional probability our prize lands behind door 4 is zero, given that the event $C = \{door\, 1, door\, 2, door\, 3\}$ has occurred.

It is often possible to intuit the conditional probability of an event, but a concrete formula will be essential for more complex scenarios. The conditional probability of an event can be calculated using some familiar terms

$$\frac{P(A \cap B)}{P(B)} = P(A|B)$$

Indeed, the idea of conditional probability can be a sticking point for new learners of probability, so it's worth doing a few examples.

Example 1.18. First, let's define a sample space - $S = \{1, 2, 3\}$ - and assign a probability of 1/3 to each outcome. Now, let's define two events - $C_1 = \{1\}$ and $C_2 = \{1, 2\}$. We shall endeavor to calculate $P(C_1|C_2)$. In other words we want to know the probability of a "1" outcome/event in our random experiment, given the event C_2 has occurred. Intuitively speaking, if we know that the outcome has landed in $C_2 = \{1, 2\}$, we may feel as if the probability of getting a 1 is increased. Indeed, the outcome of 3 is now out of the picture. Let's not lean on our intuition though. Looking at the equation for conditional probability, we shall need two quantities for our calculation - the probability of C_1 AND C_2 ($\cap$) and the probability of C_2:

The tools in the last section will prove sufficient to calculate these.

$$P(C_1 \cap C_2) = P(\{1, 2\} \cap \{1\}) = P(\{1\}) = 1/3P(C_2) = 1/3 + 1/3 = 2/3$$

Now that we have what we need, it's just a matter of plugging in the numbers to the conditional probability equation.

$$\frac{P(C_1 \cap C_2)}{P(C_2)} = P(C_1|C_2)\frac{1/3}{2/3} = \frac{1}{2}$$

The probability of C_1 given that C_2 has occurred is 1/2.

The computation isn't too hard (although it may be at first), but the interpretation of conditional probability can certainly be tricky. It's worth unpacking our last example a bit more. One way to envision the last example is to think of a probability pie (stick with me here). In the pie chart below, observe how each of the individual outcomes is assigned an equivalent amount of area. That is, each gets 1/3 of the total probability.

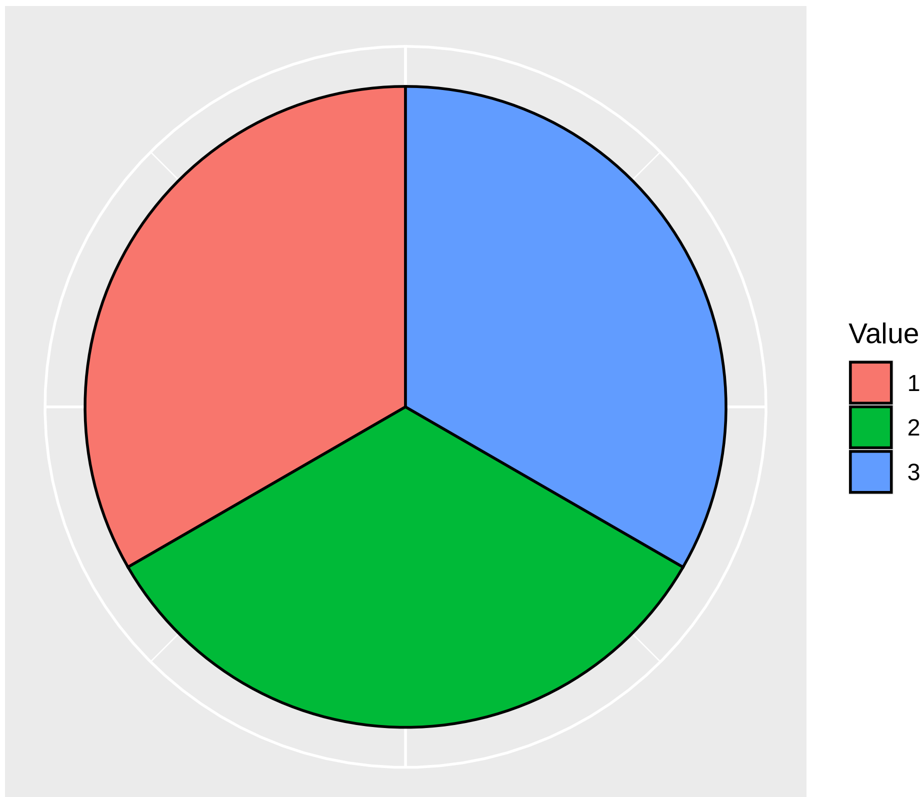

Figure 1.3: Probabilities before C2 occured

Now, we are learn that the event $C_2 = \{1, 2\}$ has occurred. Put another way, we now know that the outcome from our random experiment has landed *somewhere* in C_2. The wedge in our pie chart corresponding to outcome 3 is removed. You can see this in the pie chart below (it's now white). Observe that the two remaining outcomes - 1 or 2 - have an equal amount of probability assigned to them. One might intuit that, in the present circumstance, each of the two outcomes would occur about half the time. Indeed, the result of our calculation confirmed this. The conditional probability of getting a 1 - given C_2 occurred - is 1/2. Because C_2 occurred, the probability of C_1 increased from 1/3 to 1/2.

Example 1.19. Let's take a look at an example that will give us another view of things. Suppose we flip two coins and, as in previous examples, our sample

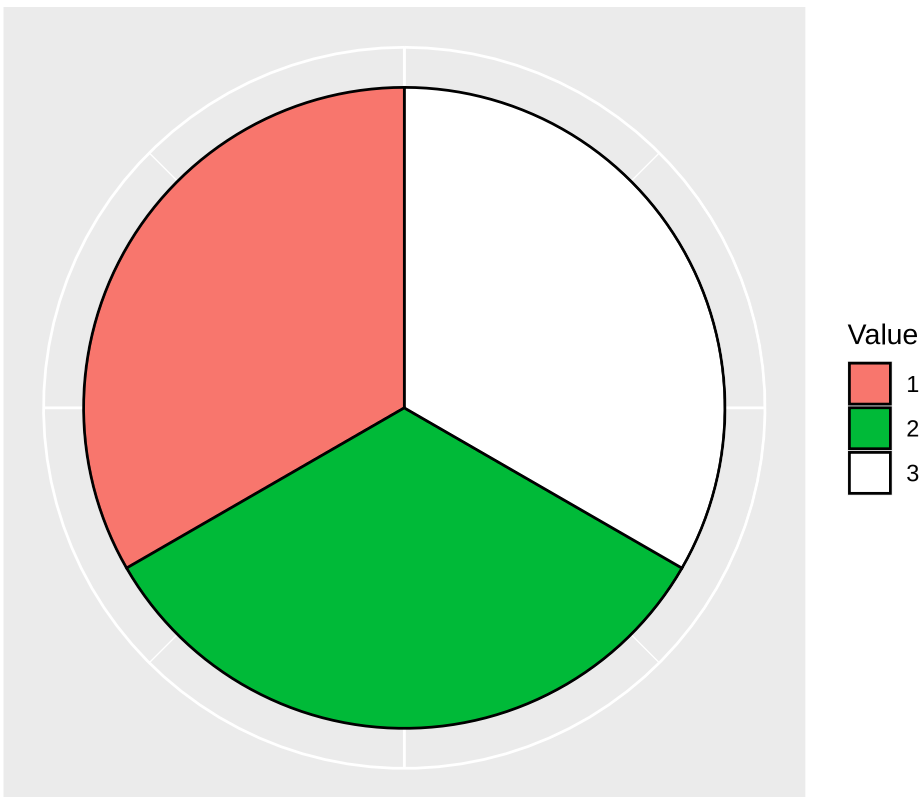

Figure 1.4: Conditional probability now that C2 has occurred

space is $S = \{TT, TH, HT, HH\}$. Like before, our probability set operator assigns to each a probability of 1/4. In this case, what is the probability of getting a head on the second flip - given we've flipped a head on the first?

Recall from our previous work, and the note at the end of the section 2.1, that flipping a head on the second coin is described by the event - $C_2 = \{TH, HH\}$. Note that here I've used C_2 to remind you that we are referring to a head on the second flip. Getting a head on the first flip is described by $C_1 = \{HT, HH\}$. The probability of each of these events is:

$$P(C_1) = P(\{HT, HH\}) = 1/4 + 1/4 = 1/2 P(C_2) = P(\{TH, HH\}) = 1/4 + 1/4 = 1/2$$

In order to calculate the conditional probability - $P(C_2|C_1)$ - we shall also need $P(C_1 \cap C_2)$ - e.g., the probability that our outcome lands in C_1 AND C_2.

$$P(C_1 \cap C_2) = P(\{HT, HH\} \cap \{TH, HH\}) = P(\{HH\}) = 1/2$$

With $P(C_1)$ and $P(C_1 \cap C_2)$, we can now calculate $P(C_2|C_1)$

$$\frac{P(C_1 \cap C_2)}{P(C_1)} = \frac{1/4}{1/2} = \frac{1}{2}$$

The probability of flipping a second head, given you've already flipped one, is 1/2.

Indeed, we already observed that the $P(C_2)$ (head on second flip) was 1/2, and, interestingly, $P(C_2|C_1)$ was also equal to 1/2. Put another way, the occurrence of a head on the first flip did not change the likelihood of a head on the second flip. In terms of probability, we say these two events are **independent**. In a more concrete manner, given two events - A and B - with each assigned the probabilities $P(A)$ and $P(B)$, the two events are independent if

$$\frac{P(A \cap B)}{P(B)} = P(A|B) = P(A)$$

The occurrence of B doesn't change the probability of A; the probability of A occurring, given B occurred, is unchanged. As I promised at the end of the last section, independence and conditional probability give us a new way of calculating P(A B), and it's hidden in our equations for conditional probability. Specifically

$$\frac{P(A \cap B)}{P(B)} = P(A|B) \tag{1.15}$$

$$P(B)\frac{P(A \cap B)}{P(B)} = P(A|B)\,P(B) \tag{1.16}$$

$$P(A \cap B) = P(A|B)\,P(B) \tag{1.17}$$

The last line above - $P(A \cap B) = P(A|B)\,P(B)$ - is known as the **law of multiplication**. It gives us another way to calculate the $P(A \cap B)$. For our purposes, this law finds it's greatest use in the case of events that are independent. Recall from above that in the case of independent events - $P(A|B) = P(A)$. Therefore the law of multiplication becomes

$$P(A \cap B) = P(A)\,P(B)$$

In other words, in the case of two independent events, we can calculate the probability of A and B ($P(A \cap B)$), by multiplying $P(A)$ and $P(B)$ together. Let's give these equations some run:

Example 1.20. Suppose we are flipping three coins *independently*. That is, assume the event we get a heads or tails on a flip is independent from getting a heads/tails on any other flip. Let's say C_1 represents the first flip event, C_2 the second, and C_3 the third. Here I'm using a shorthand because writing out the sample space would be tiresome. For each coin, assign the probability of 1/2 to the event that the coin flips heads (and 1/2 that it flips tails). What is the probability of $C_1 = \{H\}$, $C_2 = \{H\}$, and $C_3 = \{H\}$? We can use the law of multiplication for independent events

$$P(C_1)\,P(C_2)\,P(C_3) = 1/2 \times 1/2 \times 1/2 = 1/8$$

We shall get three heads about 1/8th the time. As an interesting aside, suppose you wrote out the sample space and specifically identified the events described by my shorthand for C_1, C_2, and C_3, you'd find that the intersect of the sets is $\{HHH\}$. I didn't show you the exact method to determine the intersect between three sets, but you can simply do it by comparison.

You may have also noticed that I used the law of multiplication for three events - not two as I described earlier. However, if a set of events are mutually independent, you may multiply their probabilities together to get the probability of event 1 AND event 2 AND event 3... We won't discuss the mathematical definition of mutual independence here. I'll never provide you with a task in the textbook where it isn't clear.

1.4 Extended Examples of Probability

You've learned a lot about probability to this point. Great job! Probability can be a tough subject, and when I have taught people the subject, I find that students can get quite confused (albeit it could be the teacher and not the subject). It's therefore a good idea to do some more detailed and tricky examples. In this section, we'll add a little bit of flourish to our examples to make probability a little bit more clear.

As we go through each example, I recommend attempting them yourself first. Remember: *It is okay to be wrong.* In fact, being wrong is very important to fully understanding a concept, so don't get discouraged if you can't quite answer each question.

Example 1.21. Let's start on the simple side. Suppose the probability that a person likes hot dogs is - $P(\{\textit{likes hot dogs}\}) = .25$ - and the probability a person likes broccoli is - $P(\{\textit{likes broccoli}\}) = .15$. For the sake of my sanity, let's refer to these as events A and B, respectively. Further assume that if an individual likes broccoli, then the probability they like hot dogs is .10.

1. Are the events A and B independent?
2. What is the probability that someone likes hot dogs AND broccoli?
3. Bonus question: What is the sample space of our random experiment?

We shall deal with each in turn. First, we are given that $P(A|B) = .10$. You may have missed this because I used less formal language (dang word problems!). This conditional probability does not equal $P(A) = .25$. Therefore, because $P(A|B) \neq P(A)$, the two are *dependent*. Put another way, if someone likes broccoli (event B occurred), it *lowers* the probability that they like hot dogs.

The second question refers to the law of multiplication for dependent events. Here it is again:

$$P(A \cap B) = P(A|B)\,P(B)$$

We have $P(A|B) = .10$ and $P(B) = .15$. So, plugging these numbers in we get

$$P(A|B)\,P(B) = .10 \times .15 = .015$$

The probability of liking hot dogs AND liking broccoli is just .015. Out of a thousand random experiments, you might find 15 people who like broccoli and hot dogs. The actual rate is probably much higher, but we aren't talking about real data just yet.

The third part of this question concerns the sample space. I used short hand to describe the sample space, so it's worth unpacking things for clarity. Let's represent "liking hot dogs" with the symbol LH, and "disliking hot dogs" with the symbol DH. Then let's represent "likes broccoli" with LB and "dislikes broccoli" with DB. Like the outcomes of two coin flips, each event in our sample space shall represent a combination of a hot dog preference and a broccoli preference.

$$S = \{LHLB, LHDB, DHLB, DHDB\}$$

And our event "liking hot dogs" would be

$$A = \{LHLB, LHDB\}$$

Our event liking broccoli would be

$$B = \{LHLB, DHLB\}$$

The event “liking broccoli” and “liking hot dogs” would be described as

$$A \cap B = \{LHLB\}$$

And we needed independence and the law of multiplication to calculate the probability of this event - .015. Eat your veggies!

Example 1.22. Let’s turn our attention to another classic probability example: dice! Suppose we are rolling two four-sided die, and assign a probability of 1/4 to each number for each die, respectively. Let’s look at some scenarios with increasing complexity.

1. What is the probability of rolling a 1 followed by a 1?
2. What is the probability of rolling a 1 and a 2?
3. What is the probability of the dice summing to 3?
4. What is the probability of getting a sum of 3 - given you’ve rolled a 1 on the first dice?

In the first scenario, we’ll again use a short hand to refer to the outcomes of the sample space. A_1 will refer to the first roll and A_2 will refer to the second roll. We have the event $A_1 = \{1\}$ - which can also be written as the combination of all the outcomes where the first dice is 1 - $\{1\,1, 1\,2, 1\,3, 1\,4\}$. Both are equivalent. The second way of writing it is more exact. We also have $A_2 = \{1\}$, and each event has $P(A_1) = P(A_2) = 1/4$. They both have a probability of 1/4. We also know that the two events are independent. So, using the law of multiplication:

$$P(A_1 \cap A_2) = P(A_1)P(A_2) = 1/4 \times 1/4 = 1/16$$

Indeed, had we picked any two numbers for the first and second dice the probability would be the same - $1/4 \times 1/4 = 1/16$. This fact will help us with the second question above, because we now know the probability set operator will assign a probability of 1/16 to each outcome in our two roll sample space. Our two roll sample space has $4 \times 4 = 16$ possible outcomes - so the probability has been divided equally across each outcome.

The second question asks us for the probability of rolling a 1 and a 2. In this case, we could get a 1 followed by a 2 OR a 2 followed by a 1. That is to say, we are talking about the probability of the event $A = \{1\,2, 2\,1\}$, and using the law of addition

$$P(\{1\,2, 2\,1\}) = 1/16 + 1/16 = 2/16 = 1/8$$

Our work in the last step has set us up nicely to approach the third part of the question. We shall find it useful to fully specify the sample space. The table below will help us with that endeavor. Each larger row represents the first dice roll, and embedded within each larger row are all the possible rolls of the second dice. For example, let your eyes focus on the first big row - the row corresponding to a 1 on the first die. Next scan to the right and see the row corresponding to a 1 on the second dice. The probability of getting a 1 followed by a 1 (summing to 2) is 1/16.

1st Die	2nd Die	Sum	Probability
1	1	2	1/16
	2	3	1/16
	3	4	1/16
	4	5	1/16
2	1	3	1/16
	2	4	1/16
	3	5	1/16
	4	6	1/16
3	1	4	1/16
	2	5	1/16
	3	6	1/16
	4	7	1/16
4	1	5	1/16
	2	6	1/16
	3	7	1/16
	4	8	1/16

Thanks to this table we can find the combinations that lead to a sum of three. Specifically we have $S = \{1\,2, 2\,1\}$, and each of these outcomes is 1/16, so the probability of getting an outcome that sums to three is 1/8.

Now it's time for the final boss! How shall we determine the probability of the dice summing to 3 *given* we roll a 1 on the first die? How shall we determine the independence of these two events? Well, let's look at the equation for conditional probability again

$$\frac{P(A \cap B)}{P(B)} = P(A|B)$$

We shall need the probability of getting a sum of 3 AND rolling a 1 on the first die, as well as, the probability of rolling a 1 on the first die. Furthermore, for the determination of independence we will need the probability of rolling two dice that sum to 3. Let's calculate each of these in turn.

1. Let's denote the event of rolling a 1 on the first dice as $A = \{1\,1, 1\,2, 1\,3, 1\,4\}$. Way back in the beginning of this example we said that P(A) = 1/4.
2. Let's deal with the probability of our two dice summing to 3 first. Denote this event as $B = \{1\,2, 2\,1\}$. We just calculated this probability as 1/8.
3. We can calculate $P(A \cap B)$ in a couple of ways. First we can lean on our intuition a bit. Looking at the table above, we see that there is only one way to roll a 1 on the first die AND produce a sum of three - $\{1\,2\}$. The probability of this event is 1/16. If we are having difficulty leaning on our intuition, we have learned the tools to calculate $P\{A \cap B\}$.

$$\begin{aligned} P\{A \cap B\} &= P(\{1\,1, 1\,2, 1\,3, 1\,4\} \cap \{1\,2, 2\,1\}) && (1.18) \\ &= P(\{1\,2\}) && (1.19) \\ &= 1/16 && (1.20) \end{aligned}$$

Now we can calculate the probability of B given A:

$$\frac{P(A \cap B)}{P(B)} = \frac{1/16}{1/8} = \frac{1}{4}$$

And now we are prepared to evaluate independence. The probability of our dice summing to three is $P(B) = 1/8$, and $P(A|B) = 1/4$. As such, $P(B) \neq P(A|B)$. The two events are dependent. Rolling a 1 increases our overall chances of getting two dice that sum to 3. Indeed, reflecting on this a bit more, had we rolled a 4 on the first die it would be impossible to have a sum of 3 (e.g., $P(B)$ = 0). It therefore makes sense that our two events are dependent.

Example 1.23. One more example is worth considering here. Suppose we know the probability of liking the San Francisco Giants is .10. Denote this event as A. Further suppose that we know the probability of liking meatballs is .40. Denote this event as B. What is the probability of A and B - ($P(A \cap B)$?

The answer to this question is that it is unanswerable. We need more information about their independence to go any further. If the two events were independent, we could simply multiply their probabilities together. If they were not independent, we would need $P(A|B)$ or $P(B|A)$ before applying the law of multiplication.

Chapter 2

Discrete Random Variables

As behavioral researchers, we are often studying a population of something. We might be interested in the percentage of individuals who like cats or the number of cats owned by the average individual. We might be interested in the average happiness level of a population of individuals and/or what makes them happy. The populations we study are often vaguely defined, e.g., as in the population of people who have received some experimental treatment. Here, we will begin to take these somewhat vague notions and make them concrete. It turns out that, when we are studying a population, we are really studying what statisticians call a **random variable**. In this chapter, we will pay particular attention to the **discrete random variable**.

2.1 What is a discrete random variable?

When we are in the process of conducting quantitative research (and statistics), we measure things in numbers. You may have noticed that our previous discussions of discrete sample spaces used non-numeric descriptions - e.g., $S = \{H, T\}$ in the case of a coin flip. To make these sample spaces easier to work with, we will map our discrete outcomes to numbers. For example, if we were interested in a coin flip, we wouldn't work directly with the values of H and T. We would map each outcome to a value, e.g., heads to the value of 1 and tails to the value of 0. On any given coin flip, the value of our outcome would therefore be a number. A variable like this - one that assumes a *number* as a result of some random process (e.g., a flip, a random experiment, drawing out of a hat, rolling, randomly sampling, etc.) - is called a **discrete random variable**. In this section, we'll provide some key definitions and notation for discrete random variables. This will make things more concrete and clear for you and make the interesting stuff more interesting.

The set of possible numbers a random variable can take is called the **space** of the random variable, and like the discrete sample space, these outcomes shall be either finite or countable. In general we refer to a random variable with a capital letter, typically X, and a potential *outcome* of this random variable is referred to symbolically by a lower case letter, e.g., x. For convenience, events in the space of a random variable are often referenced with a simplified set notation. Suppose we are describing the set of outcomes where our random variable X attains a particular value. We will write $X = x$. For example, $X = 1$ describes the outcomes in the space of X where our random variable attains a value of 1 - $\{1\}$. We may also use an inequality - e.g., $X \leq x$ - to reference a range of outcomes.

Because X attains a value from a random process, we'll need a concrete way of describing it's randomness. Recall in the last chapter, we discussed the probability set operator - $P()$. This operator assigns the probability to an outcome - x - on the basis of a set of rules. The set of rules for the assignment of probability is called a probability distribution, and in the case of a discrete random variable this distribution is called a **probability mass function** (pmf) - $p(x)$.

In practice, the set of rules (a.k.a, a probability mass function) is frequently described like this:

$$P(X = x) = p(x) = \begin{cases} \frac{1}{2} & x = 1 \\ \frac{1}{2} & x = 0 \end{cases}$$

Note that $P(X = x)$ and $p(x)$ mean the same thing - hence the equal sign. After the large curly-brace, you will see a row for each outcome in the space of X. In that row you will find the probability for that outcome. For example, $P(X = 1) = p(1) = 1/2$. The pmf of X has two important properties. Specifically:

1. All the outcomes x in the space of X have a probability $0 \leq p(x) \leq 1$. That is, all the outcomes in the space of X must have a probability from 0 to 1. The set of outcomes for which $p(x) > 0$ is called the **support** of X.

2. Given n outcomes in the space of X, the sum of their probabilities equals 1 - e.g., $p(x_1) + ... + p(x_n) = 1$. In other words, if we are conducting a random experiment by drawing a value from the space of X (the set of all numeric outcomes for X), the probability that it lands in S is 1. This hearkens back to (**??**) in the previous section.

In plain language these two properties mean that we can define the pmf of a random variable by taking a total of one unit of probability and spreading it across a set of outcomes (the support of X). For example, you might give .25 probability to a value of 1, .5 to a value of 2, and .25 to a value of 3. In this case, each outcome would have a probability from 0 to 1, and these values would sum to 1.

In this chapter we will dig into discrete random variables in three steps:

1. We shall start by discussing the independence of random variables - with the goal of framing an intuition for the concept.
2. We will then turn our attention to things researchers might want to know about discrete random variables (and, as you will learn, random variables more generally).
3. And finally, we will discuss some random variables that are frequently found in behavioral research.

2.2 Independence of discrete random variables

The concept of independence is found frequently within the field of statistics. As you read on, you will find that it is essential to the derivation, understanding, and validity of some of the most fundamental things we shall learn in statistics. Therefore, it is nice to have a deeper understanding of it. The goal of this section is to frame an intuition of the independence of random variables. A deeper mathematical understanding shall be saved for your inevitable foray into mathematical statistics (no doubt after being inspired by this book to master the topic).

In the last chapter, we discussed the concept of the independence of *events* in a random experiment. We shall now discuss independence as it pertains to random variables. And, as you will see, it is very similar to the concept of independence for events. Suppose we have two random variables X_1 and X_2. Then we say the two variables are independent if

$$P(X_1 = x_1 \cap X_2 = x_2) = P(X_1 = x_1)P(X_2 = x_2)$$

In other words, if two random variables are independent, the probability of $X_1 = x_1$ AND $X_2 = x_2$ is equal to the product of their respective probabilities. We've already seen examples of this. Recall in example **??**, we flipped three coins *independently*. If we mapped the outcomes of each coin to 1 and 0 for heads and tails, respectively, then the three resulting random variables are independent in the sense of (**??**).

With some algebraic manipulation of (**??**), we can see that the definition of independence for random variables is very similar to the definition of independence for events

$$\frac{P(X_1 = x_1 \cap X_2 = x_2)}{P(X_2 = x_2)} = P(X_1 = x_1)$$

And, perhaps not surprisingly (although it's okay if you are surprised), if the two random variables are not independent then

$$\frac{P(X_1 = x_1 \cap X_2 = x_2)}{P(X_2 = x_2)} = P(X_1 = x_1 | X_2 = x_2) \neq P(X_1 = x_1)$$

Put into the logic and hopefully more understandable intuition of last chapter, when two random variables are dependent, an event in the space of one variable *changes* the probability of an event in the space of another variable. This may seem a bit confusing at first, but you have already seen an example of the dependence of random variables. Recall example **??**, when we looked at the sum of two dice. In this case, let's call the sum of our two dice the random variable Z and represent the random variable of the first die by X.

On the basis of (**??**), we shall be looking to determine if

$$\frac{P(Z = z \cap X = x)}{P(X = x)} \neq P(Z = z)$$

The first thing we will need is $P(Z = z)$. The space of Z are all the possible sums of our two dice - $\{2, 3, 4, 5, 6, 7, 8\}$. Using the table we created in example **??**, and the law of addition, we can determine a probability distribution for Z (try it yourself and see if you get the same thing as below)

$$P(Z = z) = p(z) = \begin{cases} 1/16 & z = 2 \\ 2/16 & z = 3 \\ 3/16 & z = 4 \\ 4/16 & z = 5 \\ 3/16 & z = 6 \\ 2/16 & z = 7 \\ 1/16 & z = 8 \end{cases}$$

After our computation, we can see that the most probable value is 5. Because we haven't had much experience with random variables, it is also important to point out that this distribution has all our previously discussed properties of a pmf. The probabilities of all the outcomes in the space of Z sum to 1, and they are constrained between 0 and 1.

Now that we have determined the probability mass function for Z, we shall calculate $(P(Z = z|X = x))$. Hopefully the idea of independence/dependence of random variables will come into focus. Consider the case where $X = 1$ (a 1 was rolled on the first die). What does the distribution of $P(Z = z|X = 1)$ look like? To determine this we shall calculate the conditional probability of each Z - *given* our first dice roll is a 1. We did this back in example **??** for the specific value of $Z = 3$, and now we shall do it here for every value (try it yourself)

$$P(Z = z|X = 1) = p(z|x = 1) = \begin{cases} 1/4 & z = 2 \\ 1/4 & z = 3 \\ 1/4 & z = 4 \\ 1/4 & z = 5 \\ 0 & z = 6 \\ 0 & z = 7 \\ 0 & z = 8 \end{cases}$$

Do you see what happens with the event $X = 1$? The probability of Z *changes* with a roll of a 1 on the first die. For example, getting a $Z = 2$ now has a whopping 1/4 chance, compared to a paltry 1/16 before the first dice was rolled. Moreover, the probability is now spread across only four values - $\{2, 3, 4, 5\}$. That is, the support of Z - given $X = 1$ - is now 2, 3, 4, and 5. Thus, an intuitive definition of independence pivots on changes in the pmf of one random variable given the outcome of another variable.

This idea of independence shall be sufficient for our purposes, especially as we become more sophisticated in statistics and statistical analysis. As you will see, some of the most fundamental statistical concepts require an *assumption* of independence. In many cases, we shall assume that our random variables are independent to take advantage of some convenient mathematical properties associated with independence. We'll see one of these convenient properties as soon as this chapter.

2.3 What do we want to know about discrete random variables?

A new statistics student will have made it to this point and reasonably wondered: what the heck is the point of all this back story? This skepticism is understandable, welcomed, and hopefully motivating. In this section, we shall discuss some things - as statisticians and researchers - that we might want to know about random variables. In doing so, hopefully the bigger picture comes into view. This, in my humble opinion, is where things really start to get interesting. I'm hoping you feel the same.

But first, let's do a very very short review.

2.3.1 A very quick review: The summation operator

As statisticians we will need to sum a lot of things. It will therefore be useful to have an easy way to describe the operation of summation. To that end, suppose we are summing a set of numbers

$$x_1 + x_2 + x_3 + x_4 + x_5 + \ ... + x_n$$

The **summation operator** helps us write this operation more efficiently and with less ambiguity:

$$\sum_{i=1}^{n} x_i$$

The i at the bottom of the $\sum$ is referred to as the *index*. The index is how we refer to a single element in our set of x's. For example, $i = 5$ refers to x_5. The starting index is specified at the bottom of the summation operator to the right of the equal sign. In the equation above we are starting at the 1st element of our set of x's. The value at the top of the summation operator is the index of the last element to be summed. Here we have written n which is the last element of our set of n x's. In other words

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + x_4 + x_5 + ... + x_n$$

in terms of our example above. Additionally, it is important to note that n and i are not always used to indicate the last element and the index, respectively. Here we shall try to be consistent. Occasionally, to save a little time we shall write the summation operator like

$$\sum_{x}$$

This means - given all the $x's$ in a set - sum all of them. Alright, now it's time for the interesting stuff.

2.3.2 The expectation of a discrete random variable

The first thing that we shall want to know about a discrete random variable is its expectation. Given a random variable X, we refer to its expectation as $E(X)$. We shall also sometimes refer to it symbolically as μ. In other words

$$E(X) = \mu$$

The expectation is commonly referred to as the **mean** of a random variable. The expectation of a discrete random variable X with a given pmf - $p(x)$ is calculated

$$E(X) = \sum_{i=1}^{n} x_i p(x_i)$$

In words: For each x in the space of X, multiply by its corresponding probability and sum the resulting values. An example, will help us understand the expectation a bit better.

Example 2.1. Suppose we have a random variable X with five outcomes in it's space - $\{1, 2, 3, 4, 5\}$. Further assume that the pmf of X assigns $\frac{1}{5}$ probability to each outcome.

$$P(X = x) = p(x) = \begin{cases} 1/5 & x_1 = 1 \\ 1/5 & x_2 = 2 \\ 1/5 & x_3 = 3 \\ 1/5 & x_4 = 4 \\ 1/5 & x_5 = 5 \end{cases}$$

This is an example of a **discrete uniform distribution**. Given n outcomes, the discrete uniform distribution assigns $1/n$ probability to each outcome on the space of the random variable.

Now let's calculate the expectation of this random variable.

$$\begin{aligned} E(X) = \sum_{i=1}^{5} x_i p(x_i) &= x_1 p(x_1) + x_2 p(x_2) + x_3 p(x_3) + x_4 p(x_4) + 5p(x_5) && (2.1) \\ &= 1p(1) + 2p(2) + 3p(3) + 4p(4) + 5p(5) && (2.2) \\ &= (1)\frac{1}{5} + (2)\frac{1}{5} + (3)\frac{1}{5} + (4)\frac{1}{5} + (5)\frac{1}{5} && (2.3) \\ &= \frac{15}{5} = 3 && (2.4) \end{aligned}$$

We've multiplied each value in the space of our random variable X by its probability and summed the values. The result of our computation is 3 - $E(X) = 3$ - but what does that mean exactly? Well let's explore the idea of expectation further by plotting the pmf of X. In the figure below, the probability of each outcome is plotted. You can see the outcomes of the random variable labelled on the x axis, and probability of each is represented by the bar height. In this case, each outcome is equally likely, so the bars are of equal height.

Now suppose each of those bars has *mass* (a.k.a., probability mass), and further suppose we'll place our wedge under the x axis such that the mass of the plot balances perfectly. Just by looking at the plot, if we put a wedge directly under the value of 3, there would be equivalent probability mass on either side, and the plot would balance perfectly. Notice that the value of 3 was the result of

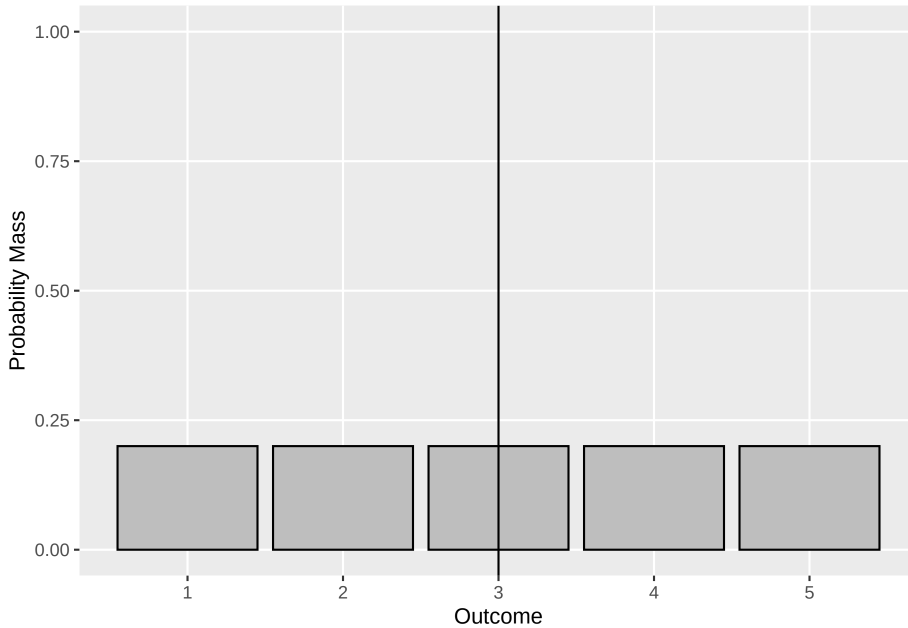

Figure 2.1: A Plot of Our PMF

our expectation computation. $E(X)$ can therefore be thought as the balancing point of the plot - the place that would perfectly balance the mass of probability for our random variable.

That said, we shall need to learn a little bit more about the expectation to proceed on our statistical adventure. First let's observe that it is possible to take the expectation of some *function* of a random variable. In other words, we may apply some function to the outcomes in the space of X and then calculate an expectation. For example, we might have

$$f(X) = 2X$$

This function takes the outcomes in the space of X and multiplies each by 2. We shall calculate the expectation of this function of our random variable by

$$E(f(X)) = \sum_{i=1}^{n} f(x_i)p(x_i)$$

Let's try it out in a couple examples.

Example 2.2. We will again use the distribution from example **??**. Here we will keep it simple (and also demonstrate a property of multiplication you may